무인 수상선의 자율주행을 위한 멀티 레이더 센싱 기술

2026.06.25

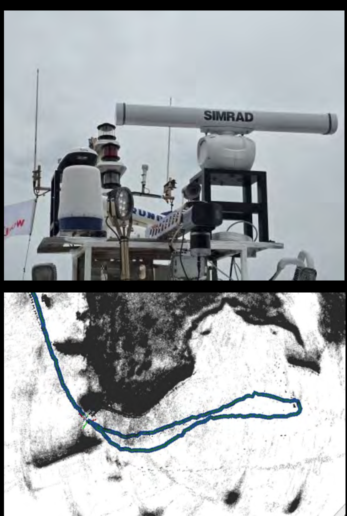
Ayoung Kim
Seoul National University
Mechanical Engineering

RPM ROBOTICS

Robust Perception in the Wild

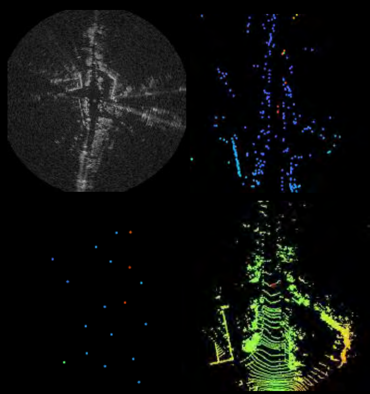
Maritime Radar SLAM

- Marine radar SLAM
- Radar for USV



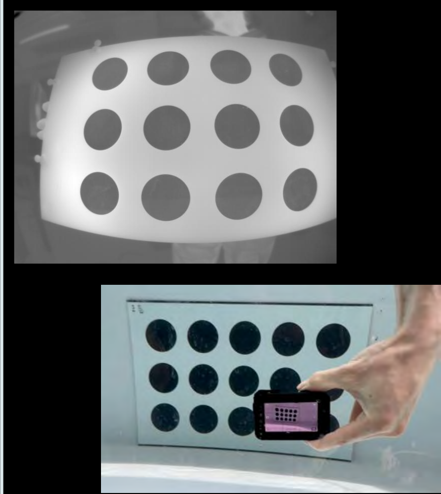
Heterogeneous Radar

- Imaging radar



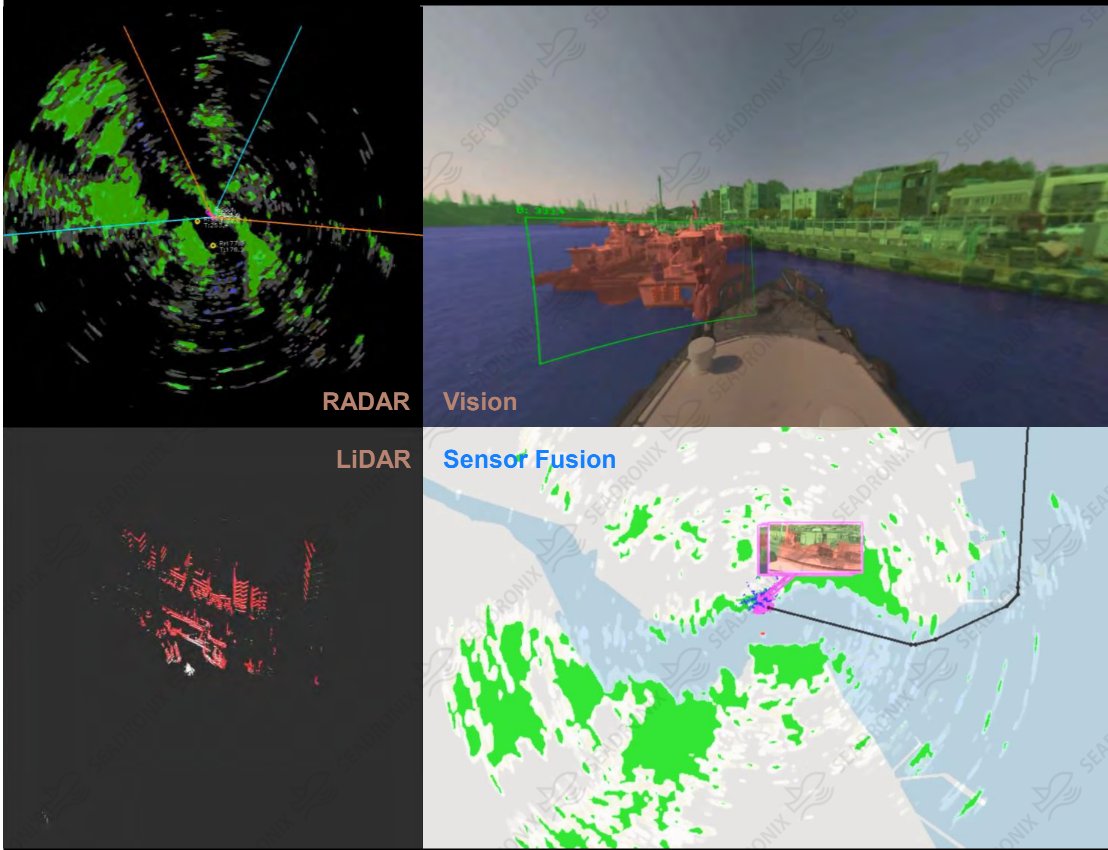
New Camera Calibration

- Conic calibration
- Thermal u/w calibration



RaPlace: Radar Place Recognition

WeBT16

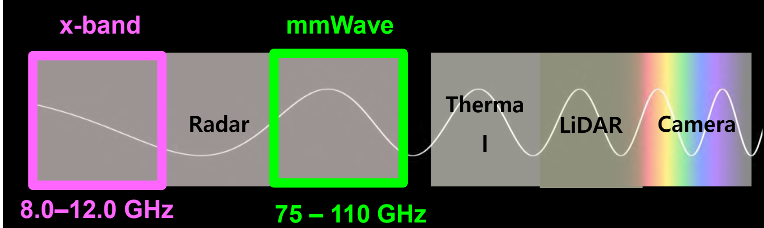


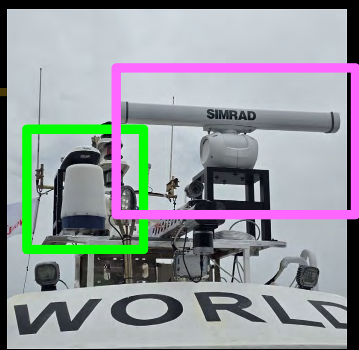
SEADRONIX

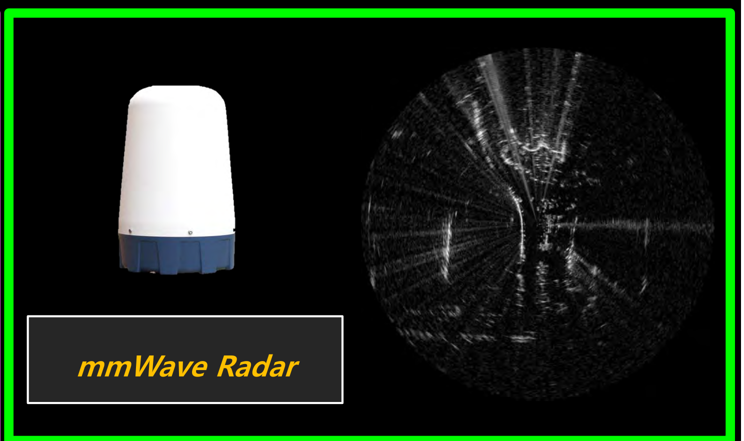
Unnamed surface vehicle

- Marine radar
- Vision
- LiDAR

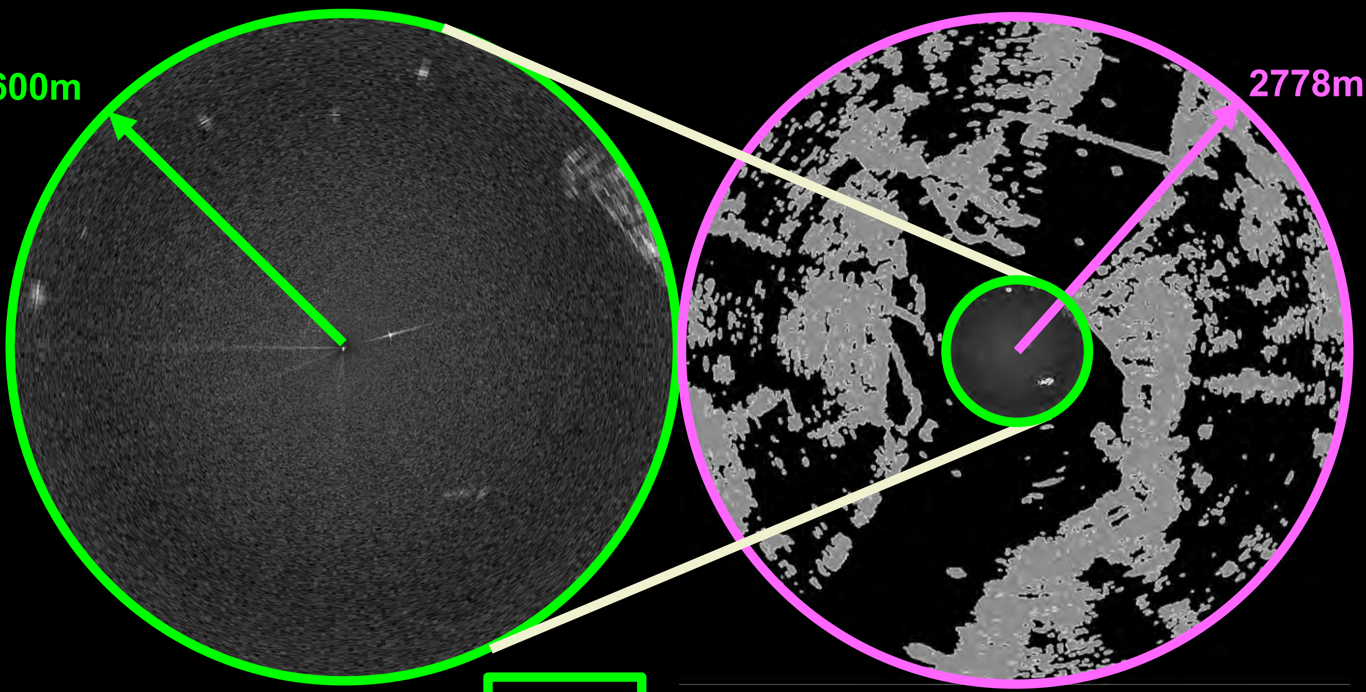
Radar Choice for Marine Robotics







Maritime SLAM and Radar Fusion



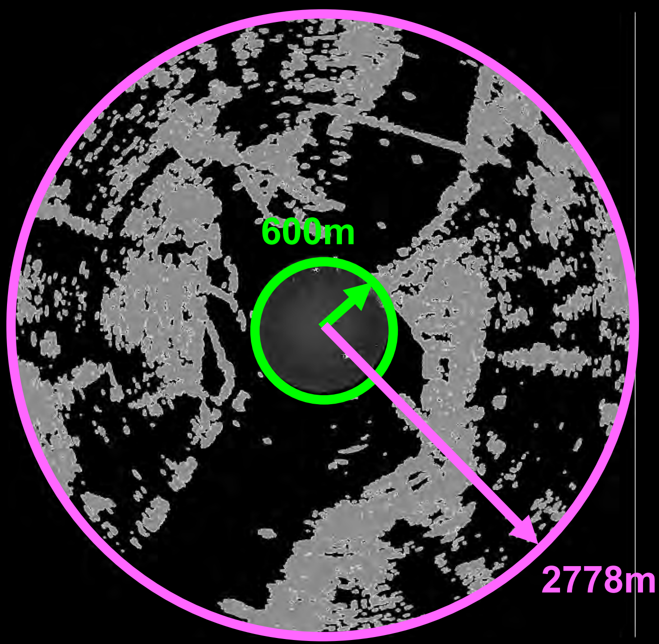
Two Imaging Radars for USV

Resolution?
Frequency?

Range?

Resolution?
Frequency?

Range?



RaPlace

Marine Radar Place Recognition
(IROS 23)

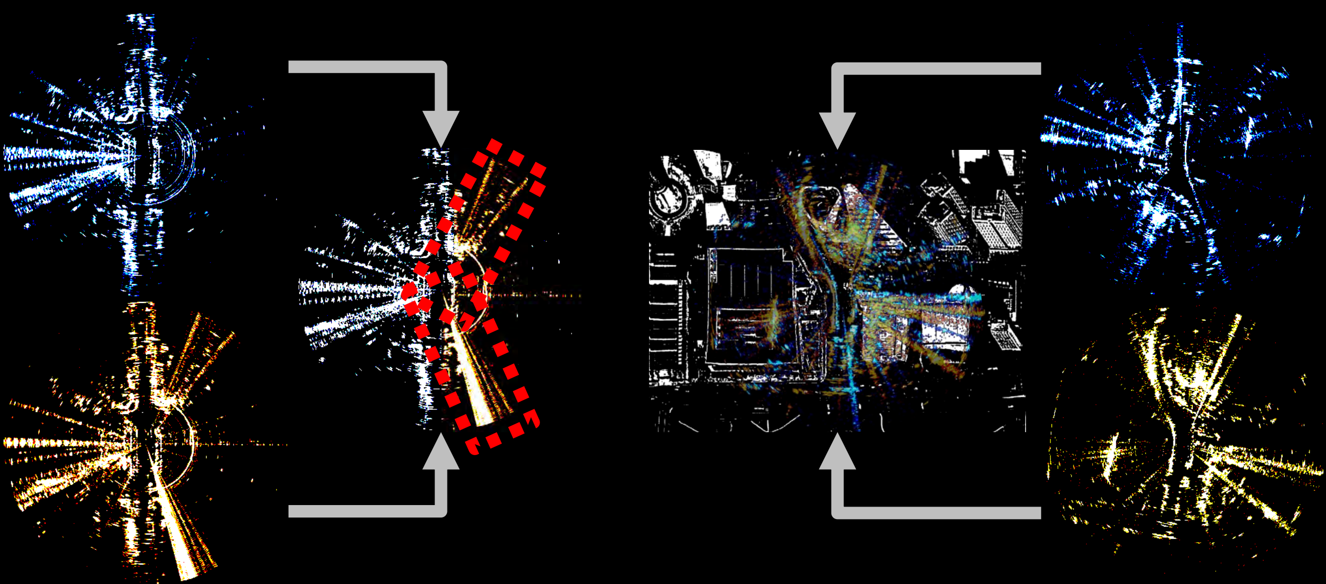
https://github.com/hyesu-jang/RaPlace

H. Jang, M. Jung and A. Kim, RaPlace: Place Recognition for Imaging Radar using Radon Transform and Mutable Threshold. IROS 2023

RaPlace: Radar Place Recognition

Radar Multipath Noise

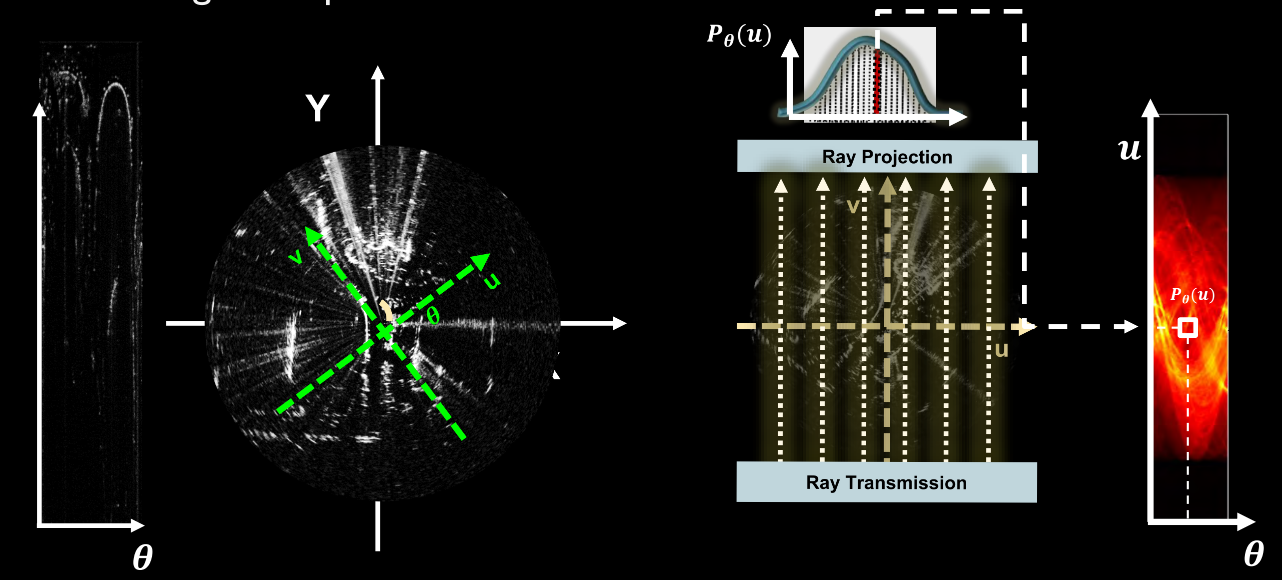
Rotation & Translation Invariance



H. Jang, M Jung, and A. Kim, RaPlace: Place Recognition for Imaging Radar using Radon Transform and Mutable Threshold, IROS 2023.

RaPlace

- Radon transform generates sinogram
  - Sinogram represents the accumulated value for each theta.

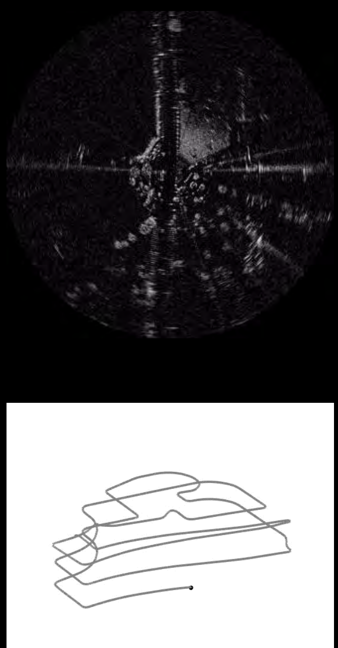


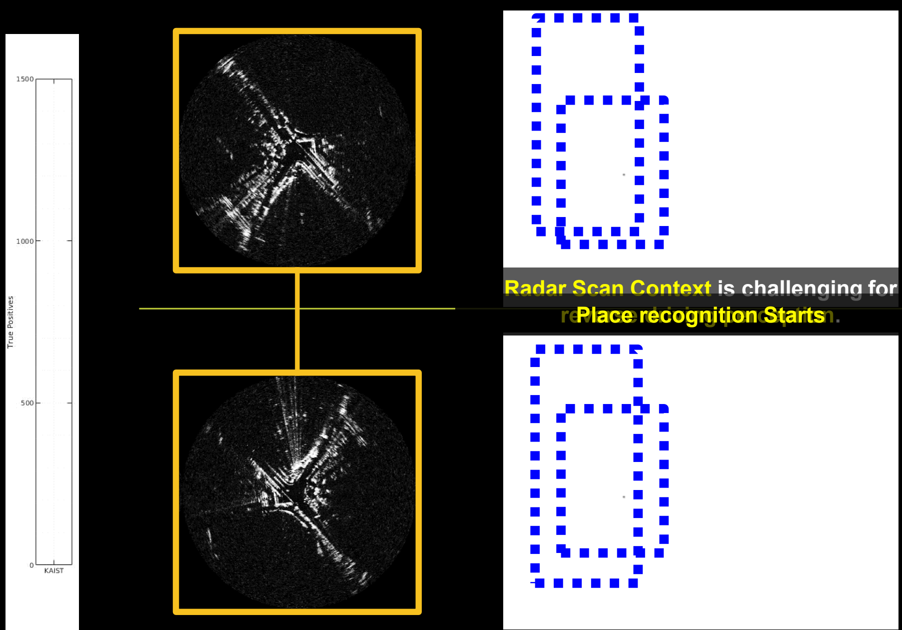
Polar *Cartesian* *Sinogram*

H. Jang, M Jung, and A. Kim, RaPlace: Place Recognition for Imaging Radar using Radon Transform and Mutable Threshold, IROS 2023.

Evaluation – MulRan KAIST

common fail matched





H. Jang, M Jung, and A. Kim, RaPlace: Place Recognition for Imaging Radar using Radon Transform and Mutable Threshold, IROS 2023.

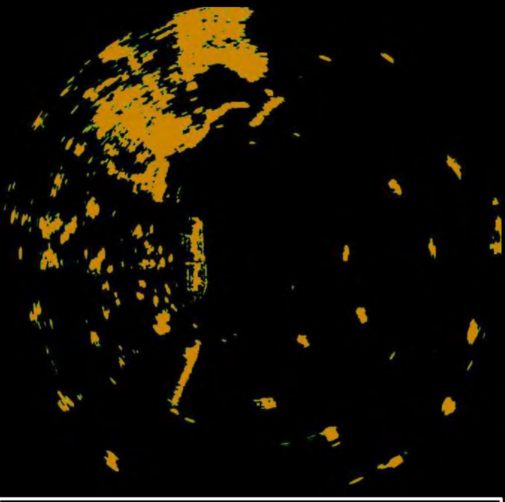
LodeStar

Imaging Radar Odometry for USV
(RAL 24 / IROS 24)

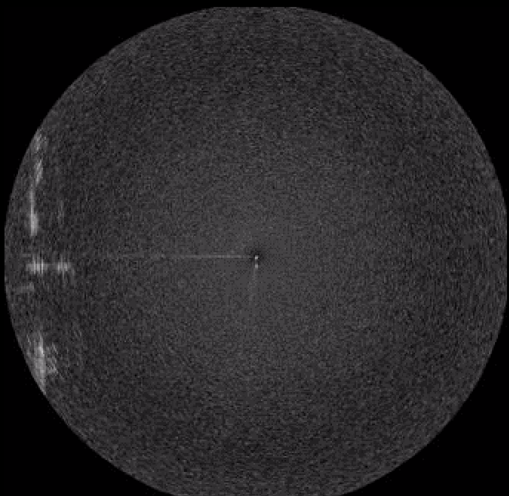
https://github.com/hyesu-jang/LodeStar

H. Jang, M. Jung, M. Jeon and Ayoung Kim, LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry RA-L 2024

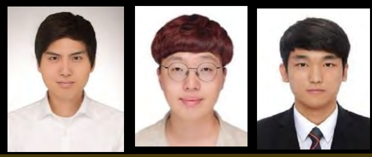
Maritime SLAM and Radar Fusion



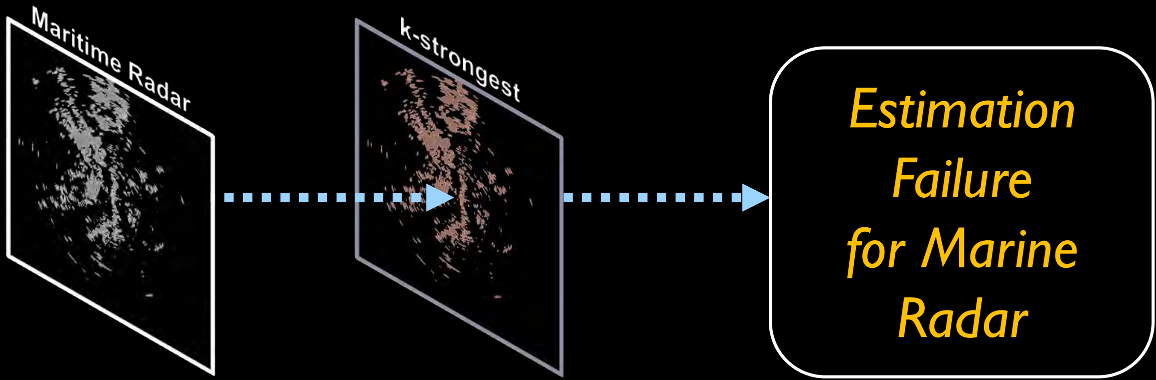
Marine Radar



mmWave Radar

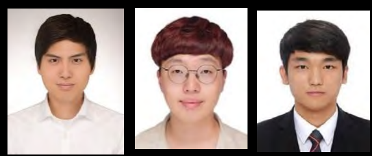


Maritime Radar Odometry

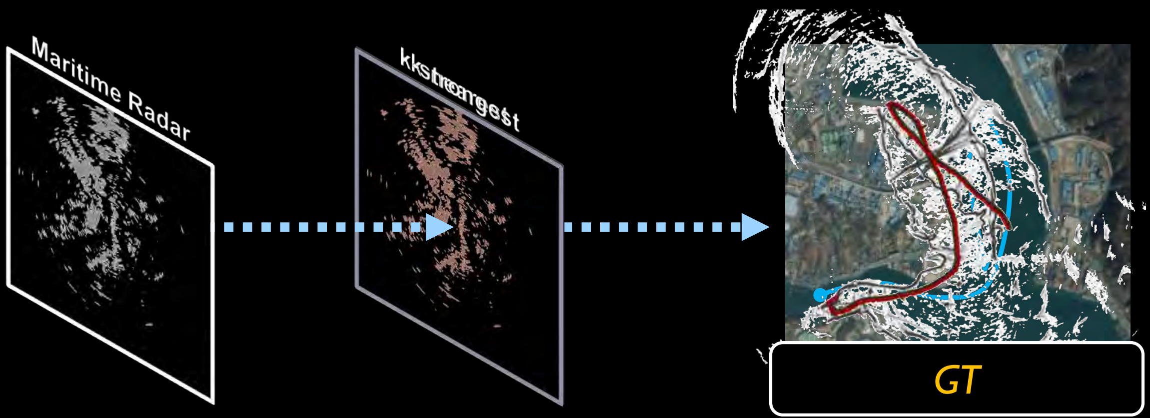


CFEAR, TRO 22

Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, Ayoung Kim "LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry." IEEE RAL 2024.



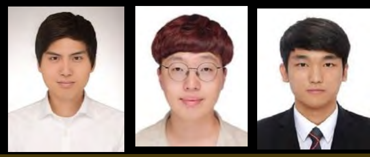
Maritime Radar Odometry



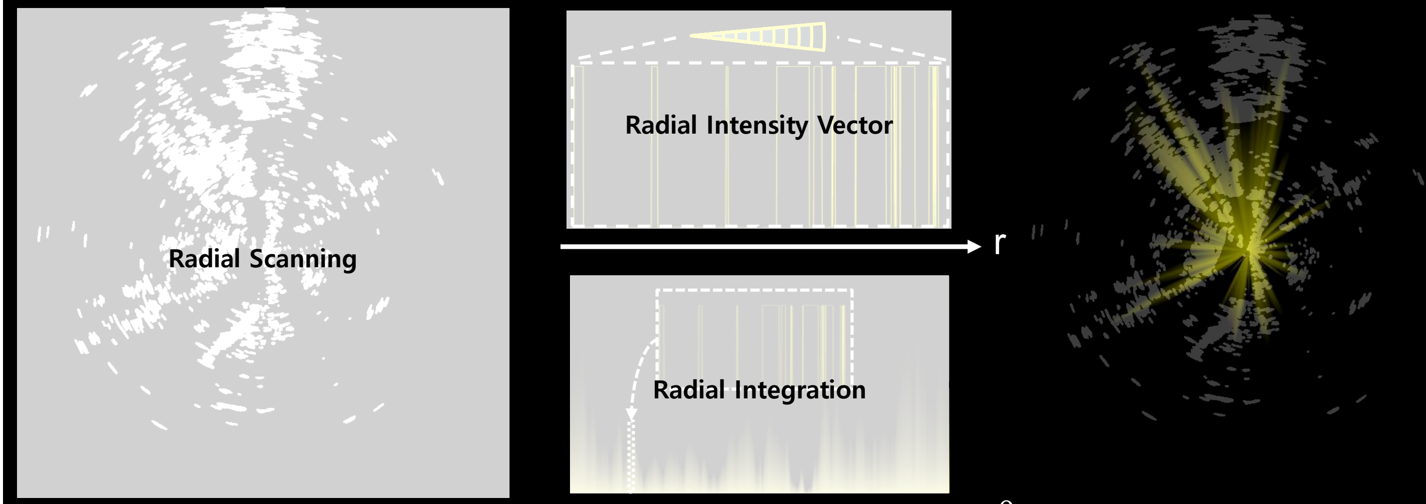
We need enhancement on orientation estimation!

Odom Failure under Large Rotation

Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, Ayoung Kim "LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry." IEEE RAL 2024.

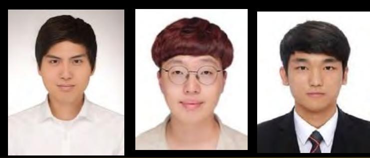


LodeStar Descriptor

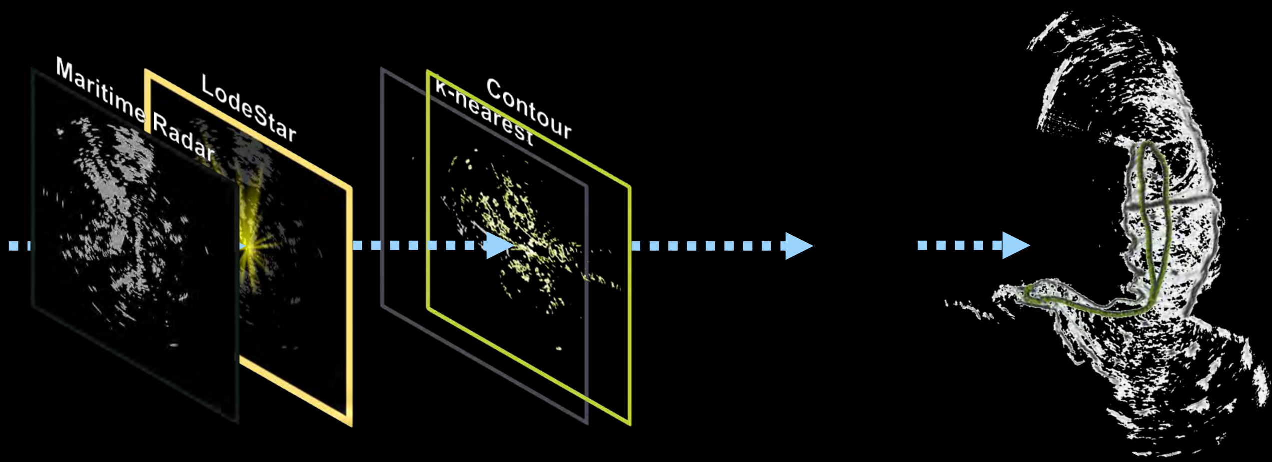


LodeStar Descriptor (Rotation Estimator)

Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, Ayoung Kim "LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry." IEEE RAL 2024.



Radar Odometry using LodeStar

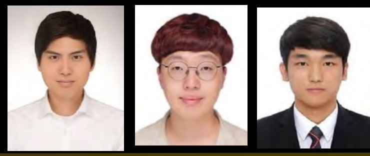


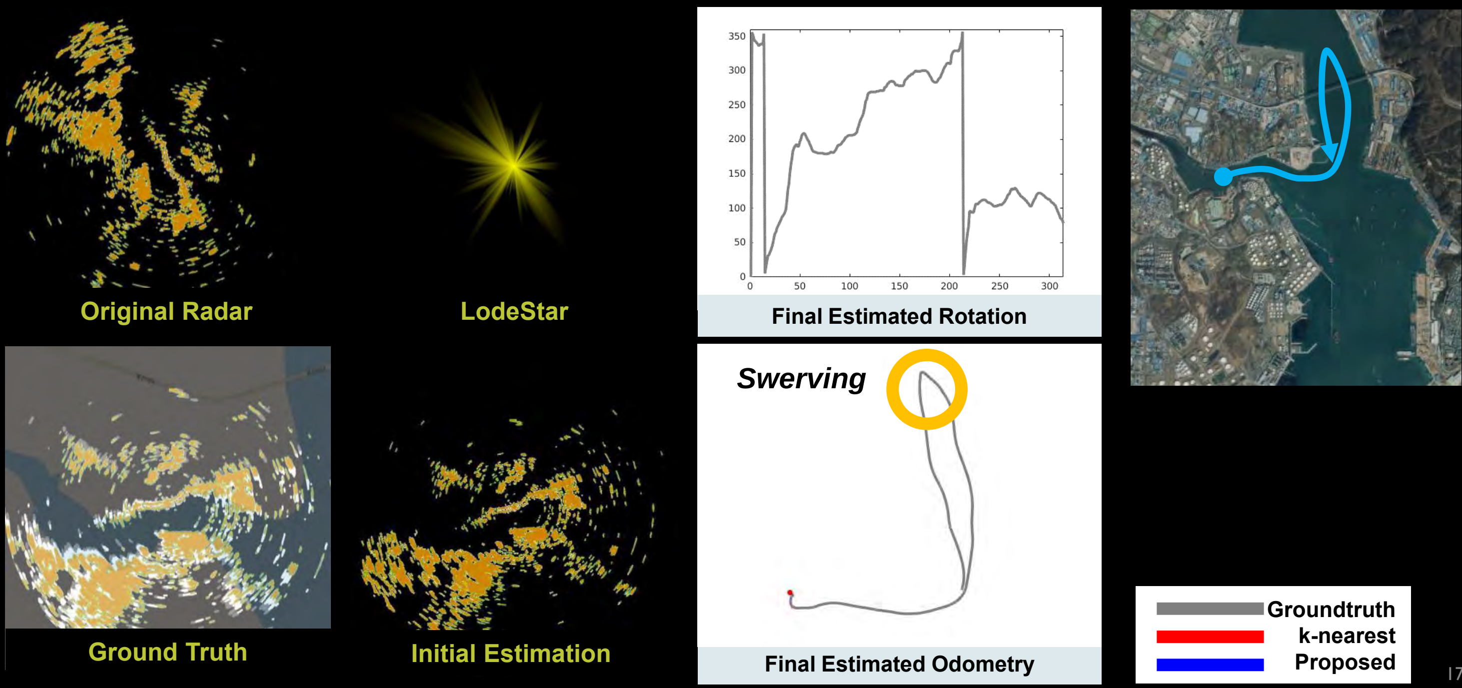
Rotation by LodeStar

Contour from pointcloud

Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, Ayoung Kim "LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry." IEEE RAL 2024.

Radar Odometry using LodeStar





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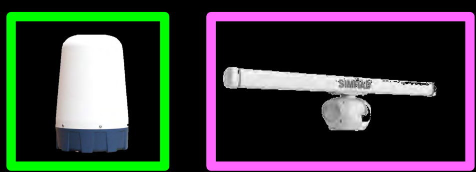
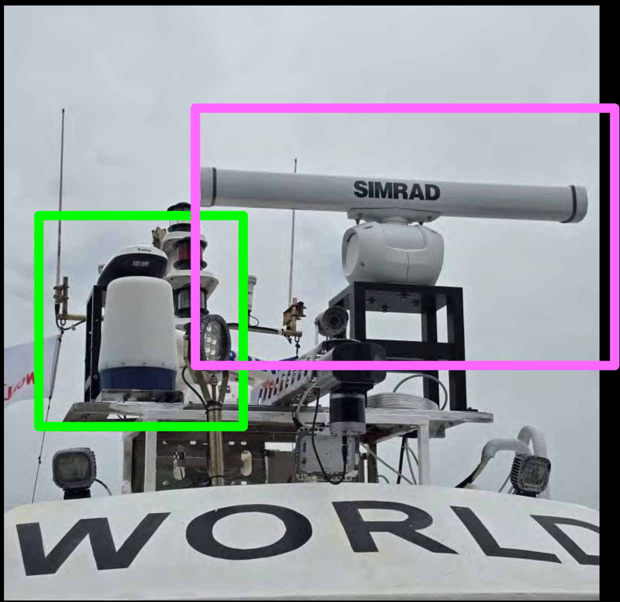
17 Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, Ayoung Kim "LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry." IEEE RAL 2024.

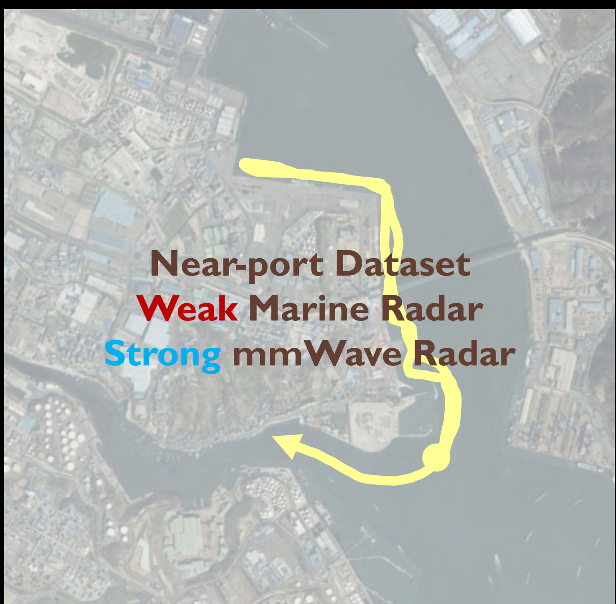
Radar SLAM

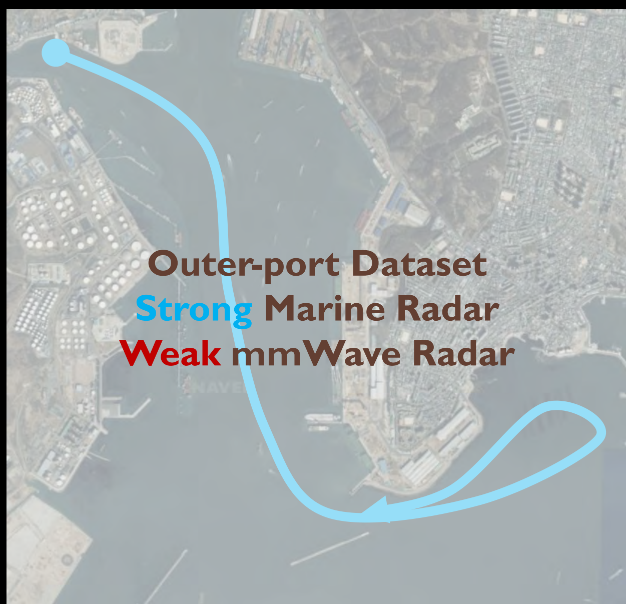


Maritime SLAM Validation



Maritime SLAM Validation





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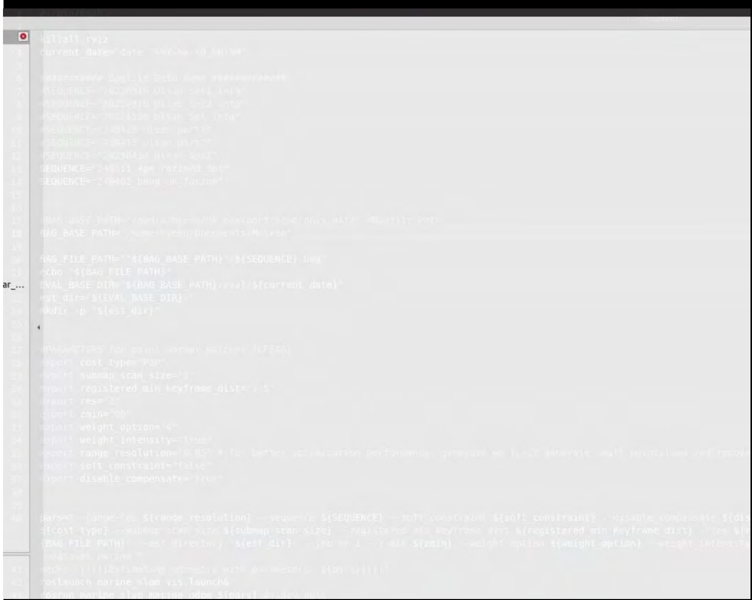
Maritime SLAM Validation

SLAM (mmWave only) near port

High noise for marine radar near shore → only mmWave radar

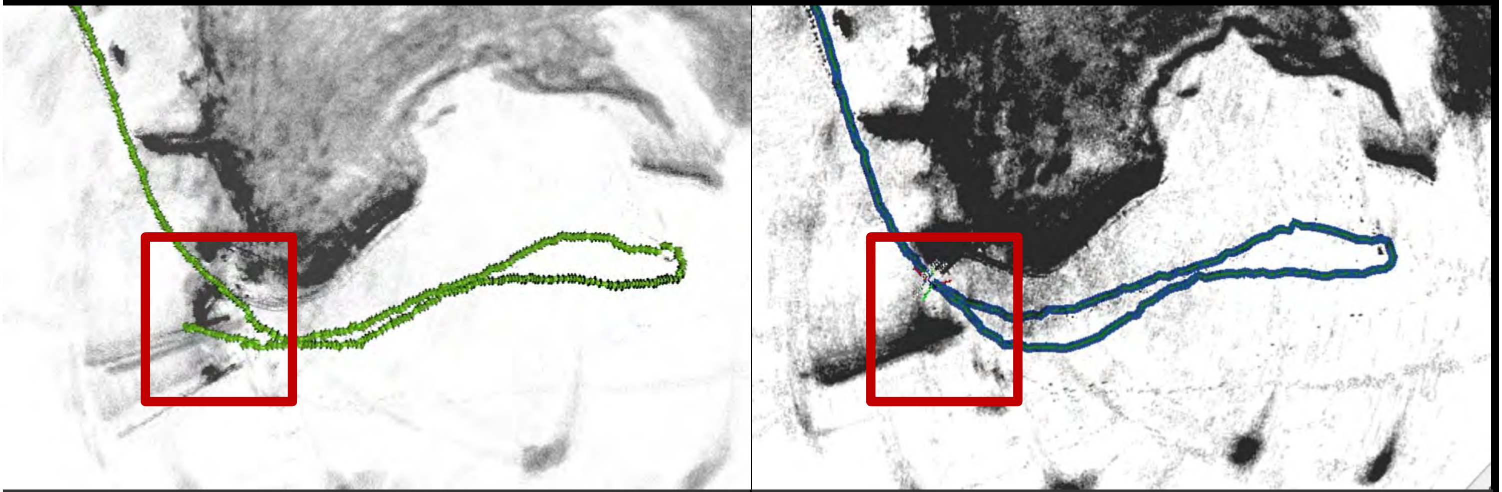
Maritime SLAM Validation



Odometry w/ Marine radar

+ PR w/ mmWave = SLAM

Maritime SLAM Validation

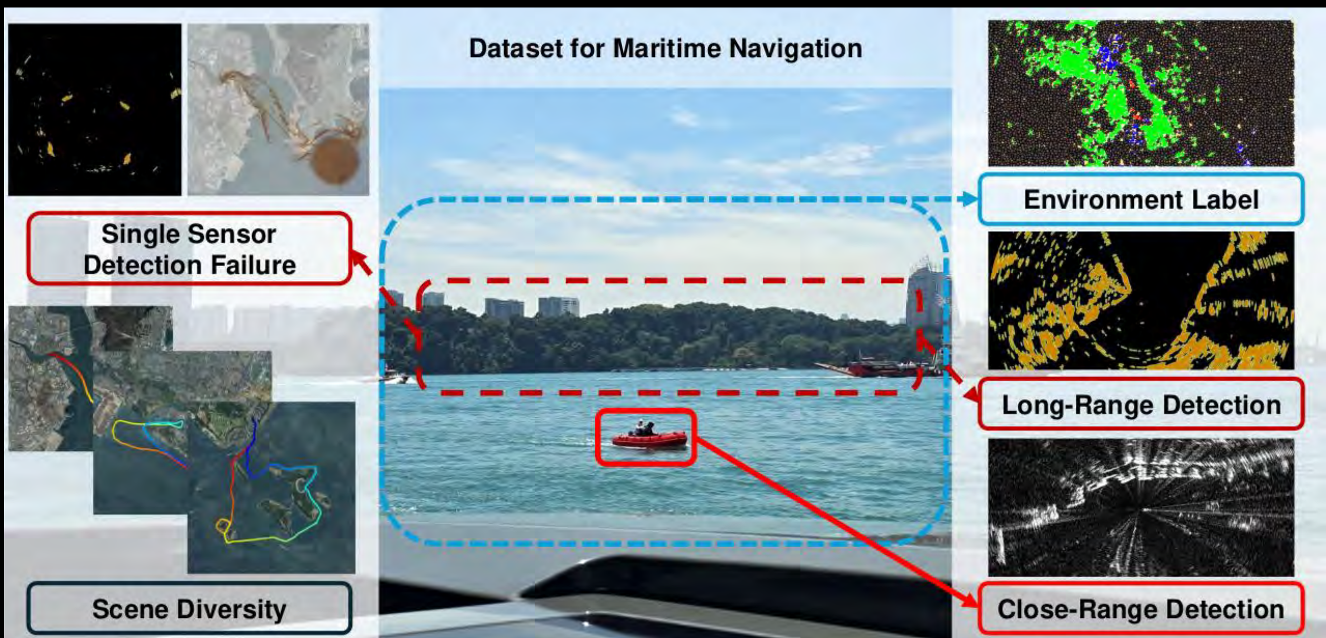


Odometry

SLAM

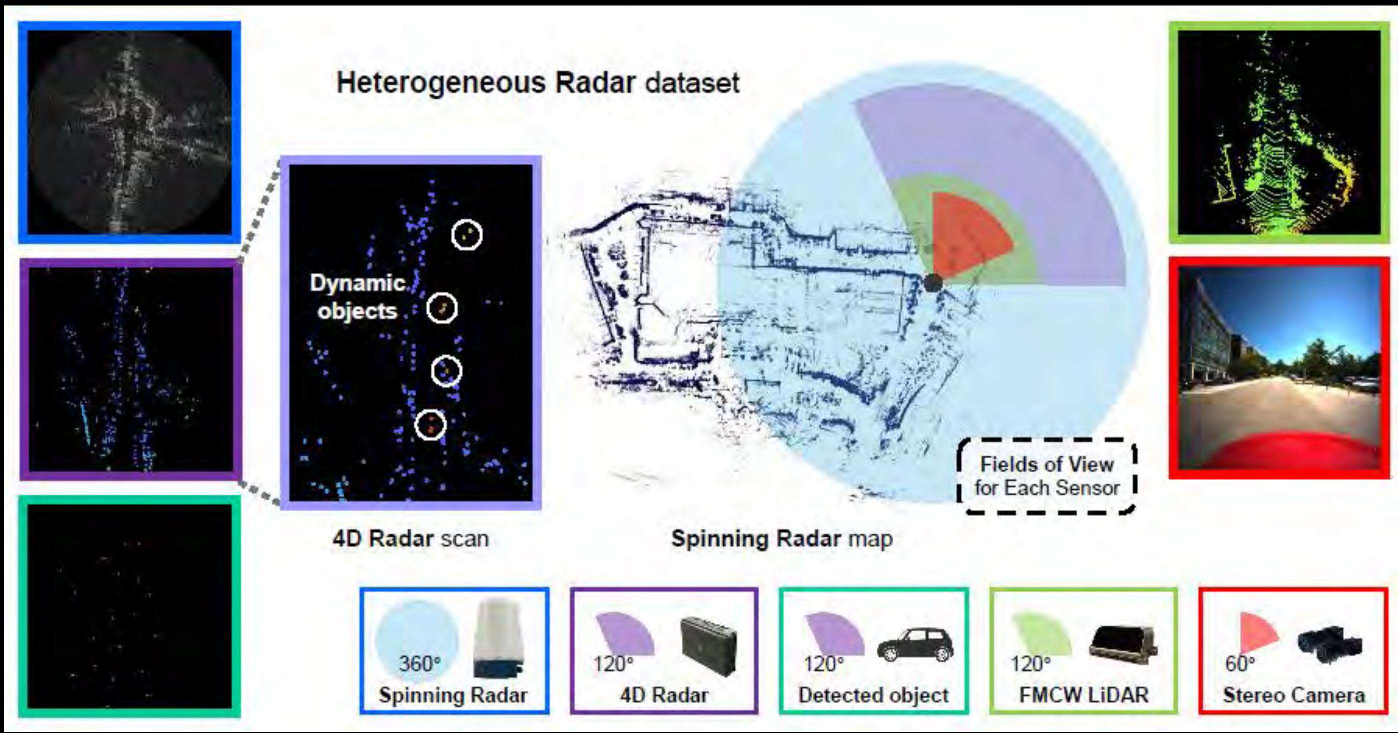
MOANA Dataset – Radar on USV

- **M**aritime **O**dometry and **A**utonomous **N**avigation **A**pplication



https://sites.google.com/view/rpmmoana/

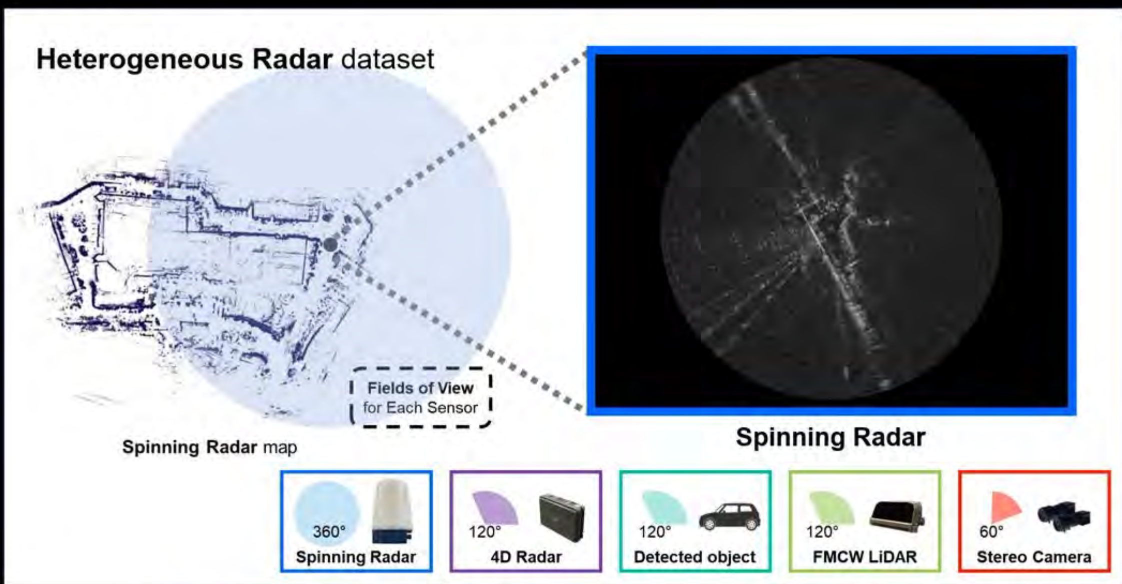
Hercules



 https://sites.google.com/view/herculesdataset

Hercules

Overview



The Spinning Radar, also known as scanning or imaging radar, provides 360° coverage and has a long detection range.

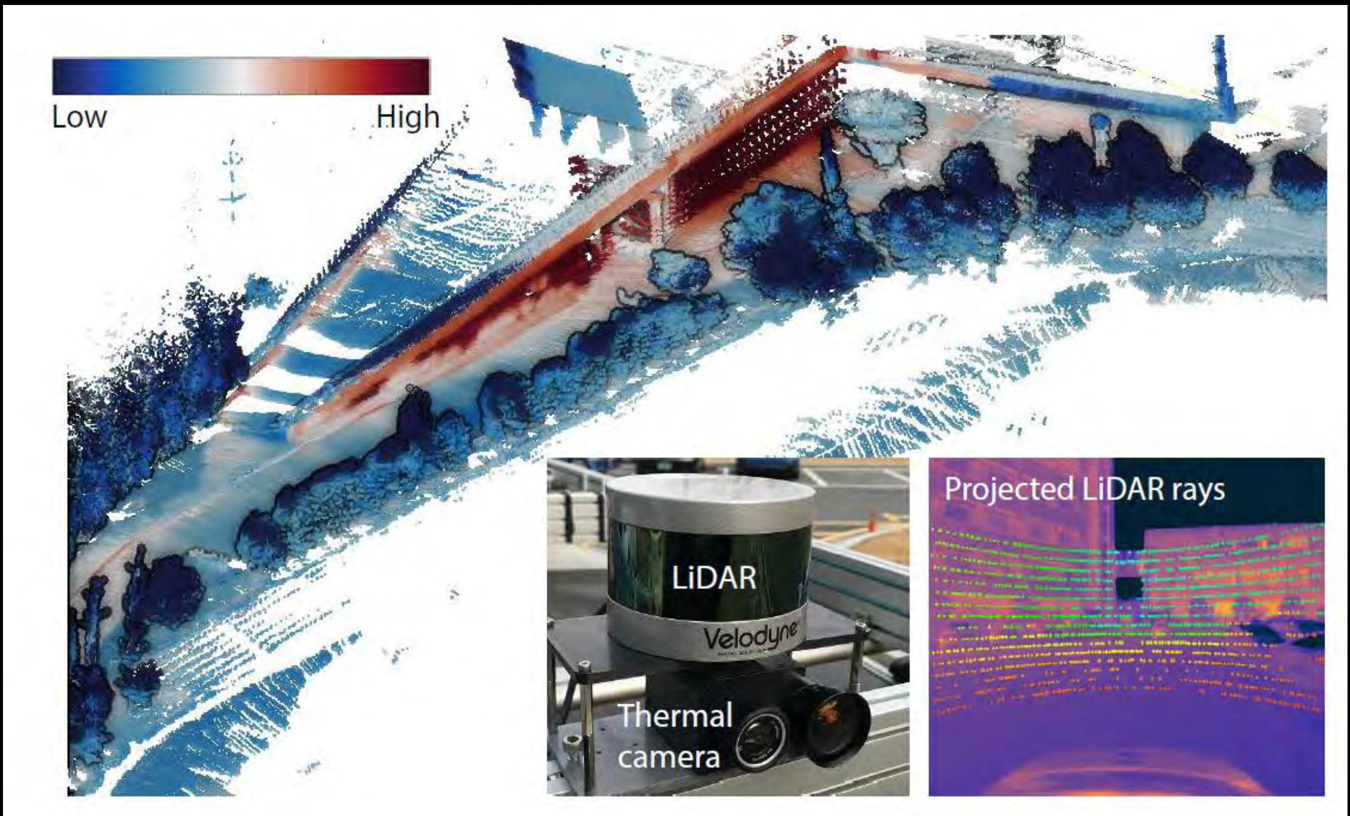
 https://sites.google.com/view/herculesdataset

Conic-based Calibration

https://github.com/ChaehyeonSong/discocal

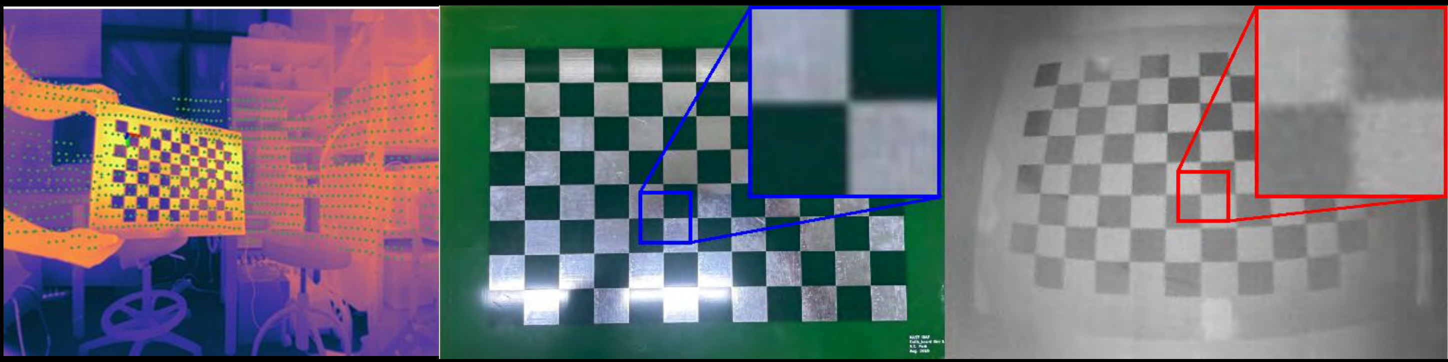
C. Song, J. Shin, M. Jeon, J. Lim and A. Kim, Unbiased Estimator for Distorted Conic in Camera Calibration. CVPR 2024 (Highlight)

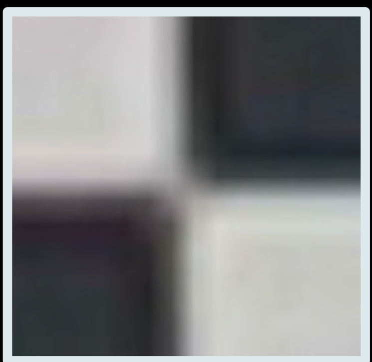
Thermal-LiDAR System



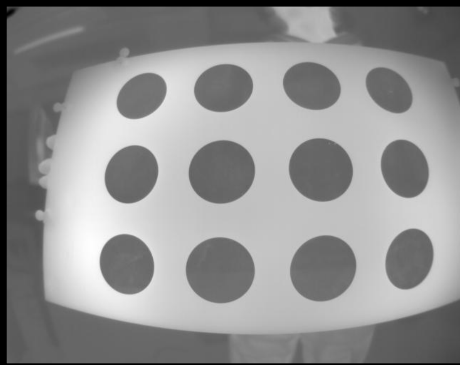
Young-Sik Shin and Ayoung Kim, Sparse Depth Enhanced Direct Thermal-infrared SLAM Beyond the Visible Spectrum. IEEE Robotics and Automation Letters (RA-L) (with IROS), 4(3):2918-2925, 2019.

How to calibration thermal cameras?



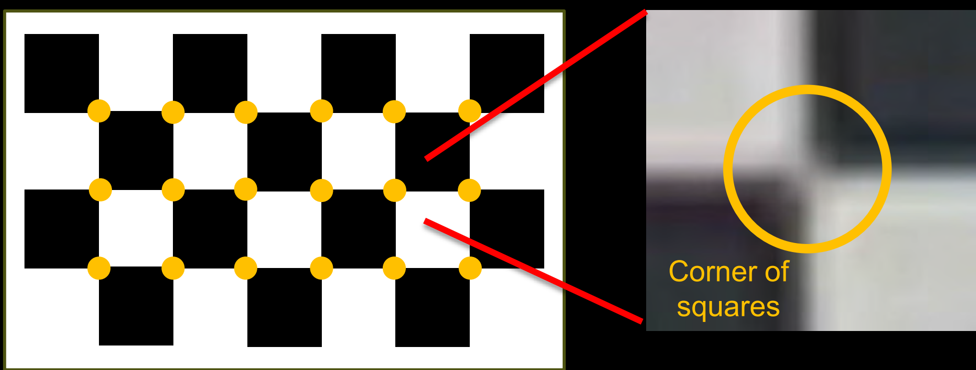


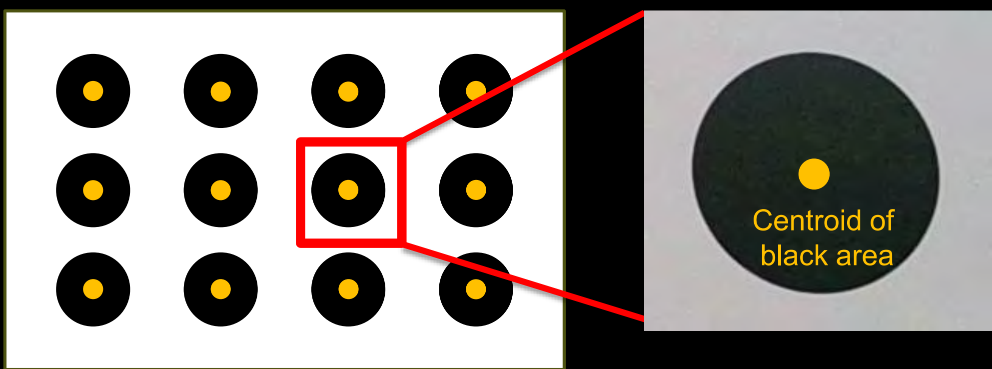
- Inherently blurry image
- Heat transfer
- Vague border lines and points



Circular pattern?

Pattern Detection: Corners vs. Circles



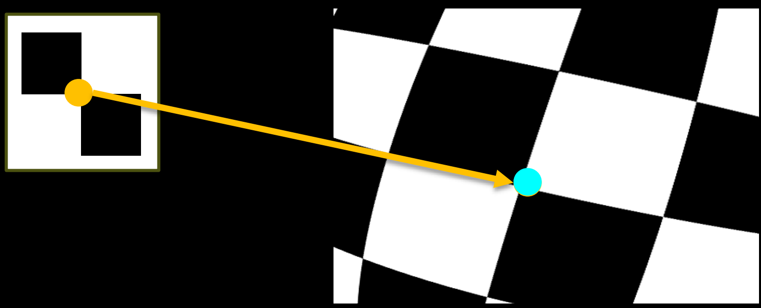


Circles are Better!

Projected Control Points

Target in 3D

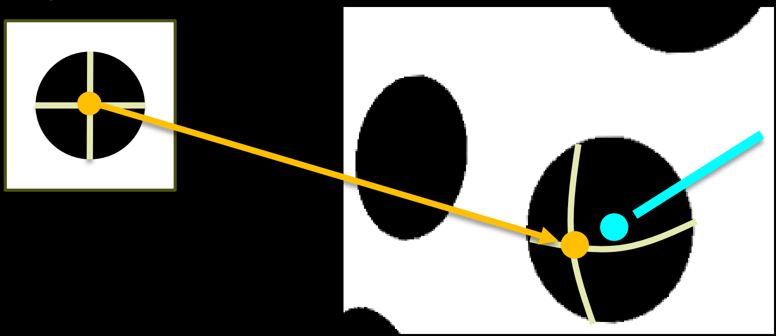
Image with distortion



Mapped point ● can also be detected even with distortion.

Target in 3D

Image with distortion



Centroid

Mapped point ● is ***NOT*** the detected centroid ●

$$K = \underset{K}{\operatorname{argmin}} \| p_{measure} - \hat{p}_{estimated}(K) \|$$

Projected Control Points

Target in 3D

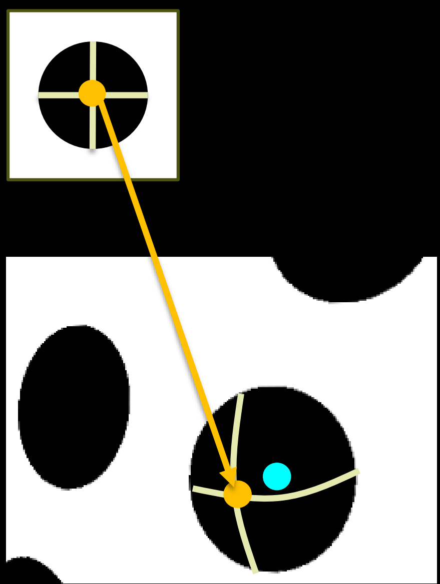
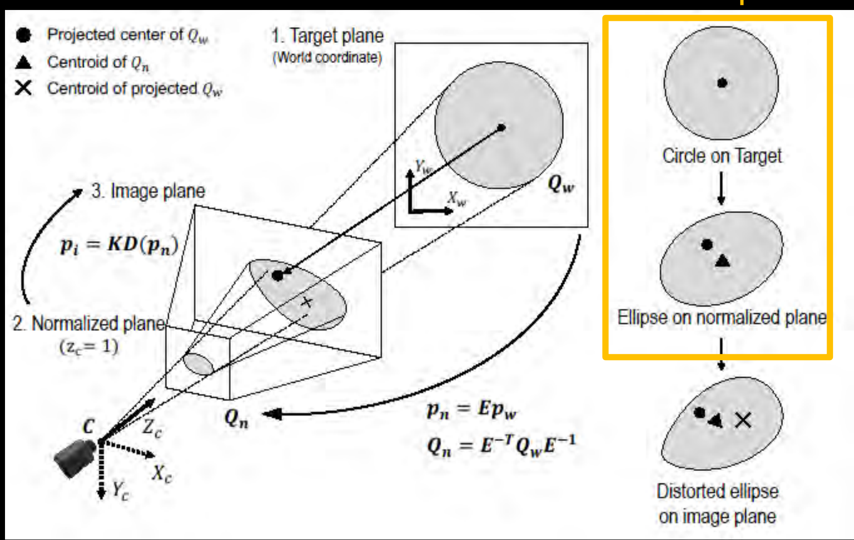


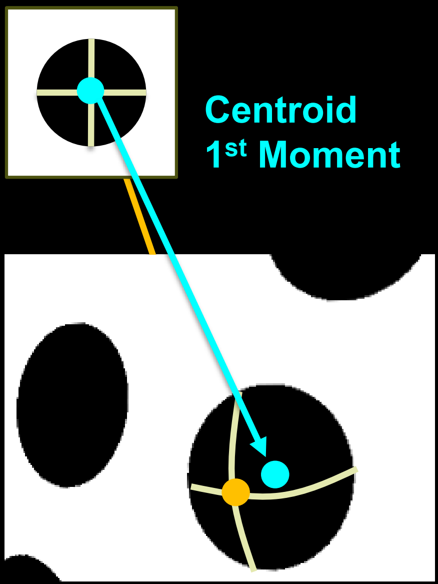
Image with distortion

Trackable to normalized plane



Conic is not conic anymore after distortion

What Should We Do?



$$K = \underset{K}{\operatorname{argmin}} \| p_{measure} - \hat{p}_{estimated}(K) \|$$

- Numerical (no analytic) solution
- Takes long time (it is okay for calibration, right?)
- Less accurate

Let's find an analytic solution

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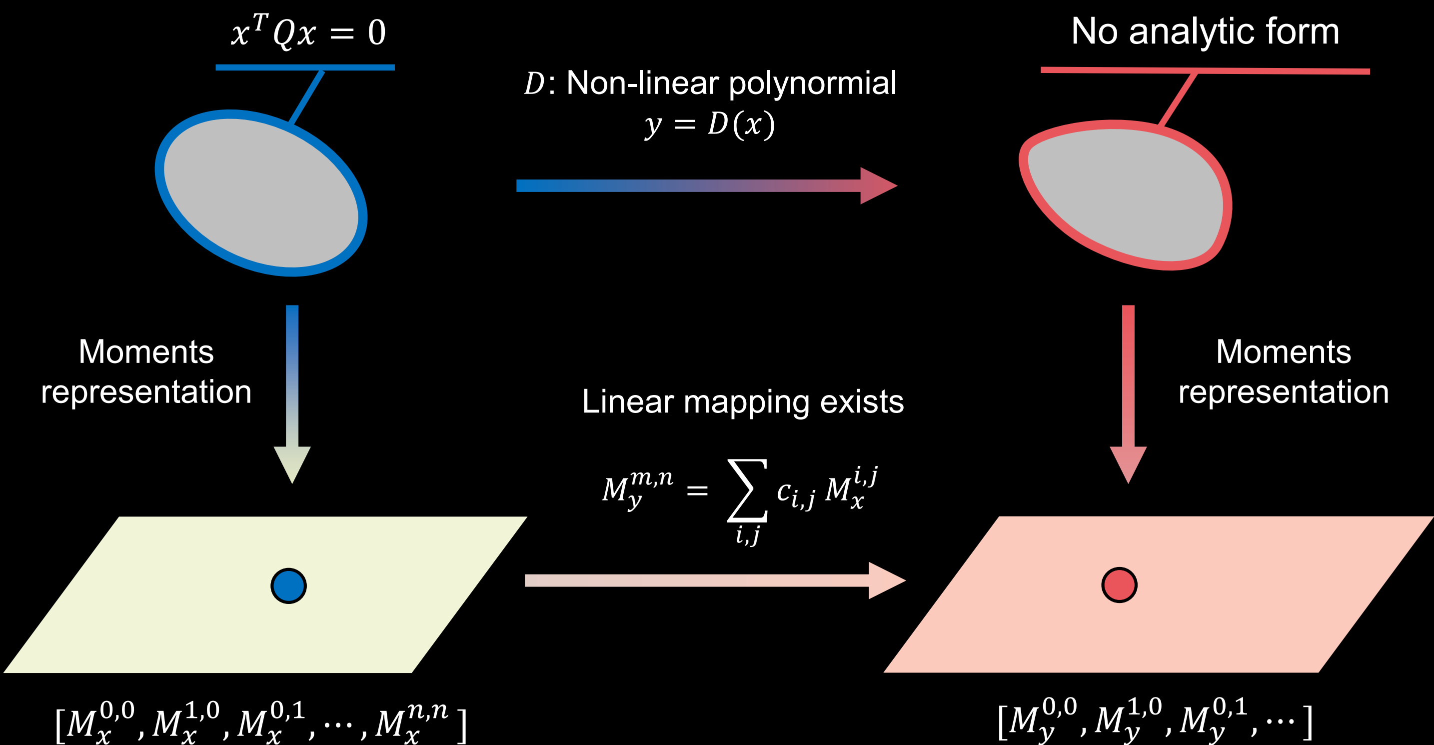
Represent Conic w/ Moments



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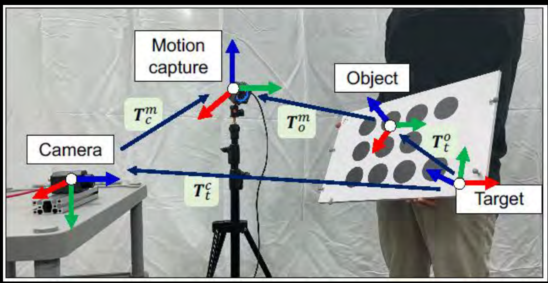
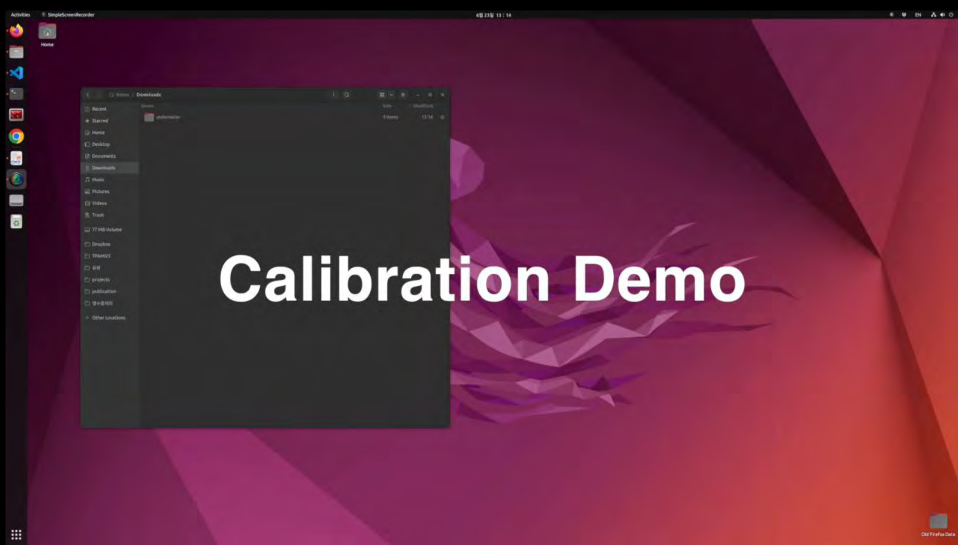
Calibration Result – 6D Pose Error



- For RGB ours par checkerboard
- For circular patters all outperformed checkerboard

	RGB		TIR	
Methods	Rotation	Translation	Rotation	Translation
checkerboard	0.32°	**2.5** mm	1.63°	16.5 mm
point-based	0.36°	4.2 mm	0.50°	3.8 mm
conic-based	0.31°	2.7 mm	0.47°	3.4 mm
numerical(1600)	0.26°	3.8 mm	**0.45°**	3.3 mm
ours	**0.24°**	**2.5** mm	**0.45°**	**3.1** mm

Underwater Camera Calibration



https://github.com/ChaehyeonSong/discocal

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Concluding Remarks

- Marine and extreme environment SLAM
- Radar fusion is very powerful for USV
  - Marine radar + imaging radar fusion

- Radar fusion for ground vehicles for all-weather SLAM
  - Imaging radars
  - 4D automotive radars
  - Doppler LiDARs

- Conic-based calibration
  - Robust and accurate calibration
  - Try out with your camera!

Acknowledgement

- This work was supported by

SEADRONIX


Hanguen Kim Ph.D.


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Dongje Lee


Woojin Ahn


Jeongmin Kim

and

Keit

IITP Institute of Information & Communications Technology Planning & Evaluation

Thank you very much !!

RPM Robotics Lab @SNU

Visit https://rpm.snu.ac.kr for papers, codes and videos!